REALIZING THE INVERSE IMAGE OF A CLOSED SUBSET OF THE COMPONENTS AS AN IND-ZARISKI QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the presentation of the inverse image of the closed subset as an intersection of open and closed subsets, the description of each piece as the spectrum of a localization at an idempotent, and the colimit construction with the resulting homeomorphism.

Editorial note. The label lemma-construct names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-construct –start 1080 –end 1095 –local-id 28009 returned RESULT=OK (L28009 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Closed subsets of the space of connected components as intersections of open and closed subsets. Topology, the lemma describing a closed subset of the space of connected components as the intersection of the open and closed subsets containing it. Supplies the presentation of the inverse image Z of the closed subset T as an intersection of open and closed subsets Z_alpha.
- Open and closed subsets of an affine scheme as spectra of localizations at idempotents. Algebra, the description of the open and closed subsets of the spectrum of a ring as the spectra of localizations at idempotents, giving local isomorphisms. Supplies the rings A_alpha with A -> A_alpha a local isomorphism and Spec(A_alpha) = Z_alpha.

Source reference. Original lemma 04PL: sources/stacks-chow/tags.txt, source lines 6119–6119. Prerequisite pointer invoked at proof line(s) 1091; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a ring. Let $X = \mathrm{Spec}(A)$. Let $T \subset \pi_0(X)$ be a closed subset. There exists a surjective ind-Zariski ring map $A \to B$ such that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ induces a homeomorphism of $\mathrm{Spec}(B)$ with the inverse image of T in X.*

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $Z \subset X$ be the inverse image of T. Then Z is the intersection $Z = \bigcap Z_\alpha$ of the open and closed subsets of X containing Z, see Topology, Lemma 04PL. For each α we have $Z_\alpha = \mathrm{Spec}(A_\alpha)$ where $A \to A_\alpha$ is a local isomorphism (a localization at an idempotent). Setting $B = \mathrm{colim}\, A_\alpha$ proves the lemma. $\square$